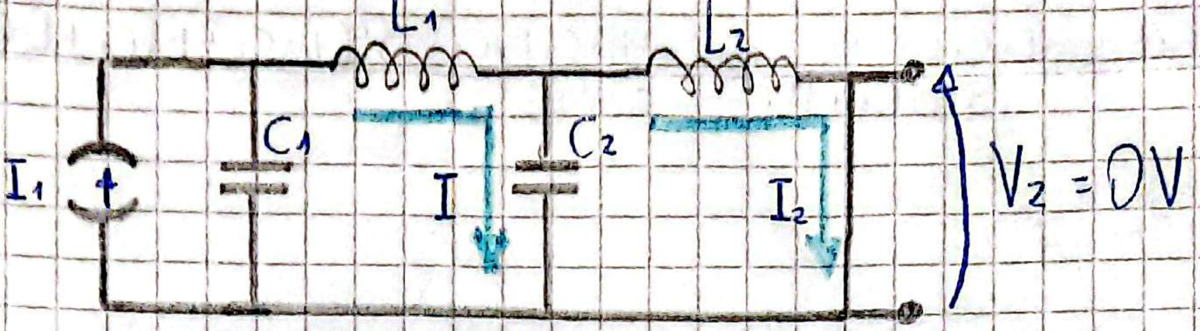
Verificación ejercicio 3



$C_1 = 1\,F \quad ; \quad L_1 = 1/6\,H$

$C_2 = 12/5\,F \quad ; \quad L_2 = 5/18\,H$

Por mallas:

$$I\cdot\left(\frac{1}{s}+\frac{1}{6}s+\frac{5}{12s}\right)-\frac{I_1}{s}-\frac{5}{12s}\cdot I_2=0 \qquad (A)$$

$$I_2\left(\frac{5}{12s}+\frac{5\cdot s}{18}\right)-\frac{5}{12s}\,I=0 \longrightarrow I=I_2\left(1+\frac{2}{3}s^2\right) \qquad (B)$$

Remplazo (B) en (A):

$$I_2\left(1+\frac{2}{3}s^2\right)\left(\frac{17+2s^2}{12s}\right)-\frac{I_1}{s}-\frac{5}{12s}\cdot I_2=0$$

$$I_2\left(\frac{17+2s^2}{12s}+\frac{(34/3)s^2+(4/3)s^4}{12s}-\frac{5}{12\cdot s}\right)=\frac{I_1}{s}$$

$$I_2\left(\frac{s^4+10s^2+9}{12s}\right)=\frac{3}{4}\cdot\frac{I_1}{s}$$

$$\frac{I_2}{I_1}=\frac{9}{(s^2+9)(s^2+1)}$$